Requirement Specification: Smart City Traffic Management

Group 2 · Reasoning and Decision Making under Uncertainty (MAIB DSC 103) · Version 2.0, 8 November 2026
Dashboard: https://smarttrafficu-mmzjikh5bg8frby3adzdc2.streamlit.app/ · Repository: github.com/Sbot-c/Smart_Traffic_U

1. Purpose and scope

This document specifies a decision-support system for urban traffic control in a 3 × 3 Downtown-Dubai-style grid (nine signalised junctions, 400 m blocks, a three-lane eastbound corridor on Street 2). Traffic lights, autonomous vehicles (AVs) and buses act as learning agents trained with Q-learning and SARSA; a decision network chooses a control strategy by weighing efficiency, safety and emissions under uncertainty; and a dashboard lets an operator explore results and run what-if simulations. The system is a research and teaching prototype run on synthetic demand, not a deployed controller. Version 2.0 adds five research extensions and corrects the comparison with a tuned actuated controller (section 12).

Stakeholders. Road authority operators (decide the control strategy), commuters and AV operators (travel time), bus operators and passengers (punctuality, crowding), pedestrians (waiting and safety), emergency services (pre-emption) and residents (emissions and health).

2. Objectives

ID	Objective	Measured by
O1	Reduce congestion and travel delay	Average delay (s/veh), throughput (veh/h), stops per vehicle
O2	Improve road safety for vehicles and pedestrians	Expected conflicts per 1,000 vehicles, jaywalks, pedestrian wait
O3	Lower emissions and their health impact	CO_2 (kg), NOx (g), idle time; decision-network health node
O4	Keep public transport reliable	Bus on-time %, bus delay, load and passengers left behind
O5	Support decisions under uncertainty	Expected utility, value of information, weight sensitivity

3. Functional requirements

ID	Requirement	Implementation
FR1	Simulate a signalised grid with time-varying demand, turning traffic, queues, spill-back and lost time	traffic_env.py: 5 s steps, 30-minute episodes, 5-minute warm-up
FR2	Model peak-hour demand, pedestrian crossings and jaywalking, emergency-vehicle pre-emption, rain and sandstorm	Scenario parameters (section 8); weather alters capacity, speed, crash risk, demand and pedestrians
FR3	Traffic-light agents choose to keep or change phase	Q-learning and SARSA signal agents, one per junction (2,250 states)
FR4	AV agents choose routes cooperatively; AVs support speed control and lane selection	RL routing (64 states); V2I speed advisory and lane selection (extensions)
FR5	Bus agents request priority or hold for schedule adherence and load balancing	RL bus agents (144 states): go / priority / hold 20 s; load KPIs
FR6	Every controller obeys a safety shield	Minimum green, pedestrian clearance, maximum green, EV pre-emption, TSP; optional pedestrian max-wait guard
FR7	Compare against fixed-time (Webster) and actuated baselines	Four policy packages; signal × routing ablation; actuated tuned on validation runs
FR8	Decision network with chance, decision and utility nodes and explicit dependencies	Weather, density, pedestrian flow → congestion, collision risk, emissions → health impact; 8 decision pairs; 3 utilities
FR9	Evaluate seven scenarios and report KPIs with uncertainty	8 held-out test episodes per result, mean ± s.d., paired tests
FR10	Interactive dashboard	Streamlit, 8 views incl. live replay, decision network, KPIs, policy comparison, Simulation Lab, research extensions
FR11	Export data and results	CSV files (section 9); downloadable from the dashboard

4. Non-functional requirements

ID	Requirement	Target / evidence
NFR1	Reproducibility	Fixed seeds: training, validation 800,000+, test 900,000+; common random numbers across policies; all data regenerated by scripts
NFR2	Honest evaluation	Model choice on validation seeds only; test seeds never used to choose; robustness over three re-trainings
NFR3	Performance	About 0.06 s per 30-minute episode; full experiments in minutes on a CPU; Lab re-runs four policies in about 1 s
NFR4	Explainability	Tabular Q-values, logged decisions and shield overrides, influence diagram with readable probabilities
NFR5	Usability	Dark operations-centre dashboard, replay with play / pause / scrubber, linked sliders, plain-language captions
NFR6	Portability	Python 3, pip requirements, Streamlit Cloud deployment; Colab notebook for the core pipeline

5. System architecture

Module	Responsibility
config.py	Network geometry, demand, weather, emission factors, scenario definitions
traffic_env.py	Mesoscopic simulator: arrivals, queues, discharge, signals, shield, pedestrians, buses, emergency vehicles, KPIs, extension switches
agents.py	Fixed-time (Webster), actuated, Q-learning and SARSA agents for signals, AV routing and buses; multi-agent system
experiments.py · run_experiments.py	Training, model selection, evaluation, ablations, data export
decision_network.py	Influence diagram, CPT learning, expected utility, value of information
course_methods.py · robustness.py	AHP, TOPSIS, uncertainty criteria, statistics; re-training robustness
extensions.py	Five research-extension studies (section 12)
app.py · viz.py · sim_view.py · lab.py	Streamlit dashboard, Plotly figures, animated replay, Simulation Lab

6. KPI definitions

KPI	Definition
Average delay (s/veh)	Time lost against free-flow travel, per vehicle, after the 5-minute warm-up
Throughput (veh/h)	Vehicles completing their trip per hour
Expected conflicts / 1,000 veh	Red-light running in the dilemma zone + rear-end exposure from stops + jaywalking, weighted by weather risk
Safety score	100 / (1 + conflicts per 1,000 / 10)
CO_2, NOx, idle time	Per-vehicle-type factors per km, per second idling and per stop
Pedestrian wait, jaywalks	Mean wait to cross; crossings on red after patience runs out (risk rises beyond 30 s)
Bus on-time %, load, left behind	Arrivals within schedule tolerance; load at each stop; passengers unable to board a full bus

7. Agent specification

Agent	State	Actions	Reward (minimised)
Traffic light (×9)	Phase, green time, queue on green and red, arrivals, pedestrian or spill-back flag	Keep / change phase every 5 s	Delay + 0.5 × pedestrian delay + 300 s per expected conflict + CO_2 term
Autonomous vehicle	Moves left, congestion of the two candidate links	Next link; speed advisory; lane selection	Travel time + delay imposed on others
Bus	Schedule deviation, load, headway, signal ahead, cross queue	Go, request priority, hold 20 s	Passenger-minutes + schedule deviation + priority cost to cross traffic

Training: 360 pre-training episodes across all scenarios with randomised weather, pedestrians and demand, then per-scenario adaptation; the general or adapted table is chosen per scenario on validation runs.

8. Scenarios

Scenario	Key parameters
Normal flow	Demand 1.0, clear weather, bus every 6 min, 3 emergency calls per 30 min
Rush hour	Demand 1.3 with a peak ramp to about 1.5, eastbound corridor +15 %, pedestrians 1.3, bus passengers 1.8
Incident / blockage	Crash before J5 eastbound, minutes 5-25, capacity −75 %, speed −60 %, 6 emergency calls
Rain	Capacity −6 %, speed −10 %, crash risk ×1.71, demand ×0.95, pedestrians ×0.60, 4 emergency calls
Sandstorm	Capacity −12 %, speed −20 %, crash risk ×2.0, demand ×0.85, pedestrians ×0.30, 4 emergency calls
Public event surge	Arena lets out at minute 8: surge of cars and pedestrians at three junctions, bus passengers ×1.5
Bus-priority corridor	Dedicated bus lane on Street 2, bus every 3 min, passengers ×2.0

9. Data requirements

Main file *smart_traffic_simulation_log.csv* (15,120 rows: scenario × policy × junction × 30 s window) with scenario, weather, policy, junction, green share, phase changes, queues per approach, arrivals, departures, vehicle delay, pedestrians waiting and their delay, jaywalks, expected conflicts, CO_2, NOx, stops, spill-back blocks, emergency-vehicle and bus-priority flags, shield overrides and reward. **Supporting files:** kpi_results.csv (per episode), decision_network_samples.csv (11,664 rows), learning_curves.csv, ablation_results.csv, training_robustness.csv, agent_decisions.csv, ped_weight_tradeoff.csv, scenario_parameters.csv, and ext_*.csv for the research extensions.

10. Decision network

Chance nodes: Weather (clear, rain, sandstorm), Traffic density and Pedestrian flow (low, medium, high), Congestion, Collision risk, Emissions and Health impact. Decision nodes: Signal timing (fixed, actuated, Q-learning, SARSA) and Routing (static, cooperative RL). Utility nodes: efficiency, safety and environment, combined with weights 0.40 / 0.35 / 0.25 (adjustable). Conditional probabilities are estimated from simulated samples with Laplace smoothing. Outputs: expected utility of all eight decision pairs, the best decision per condition, value of information and weight sensitivity. With default weights the network recommends actuated signals + cooperative routing in 6 of 7 scenarios (rain: actuated + static routes).

11. Dashboard requirements

View	Must show
City Overview	Live replay with vehicles, signals, queues and counters; KPI tiles against fixed-time; network map
RL Simulation	Side-by-side replay of two policies on the same arrivals; agent decisions and Q-values
Decision Network Visualizer	Influence diagram, expected utility of every decision pair, outcome probabilities, value of information
Scenario Explorer	The seven scenarios, their parameters and KPI distributions
KPI & Trade-off Analysis	Improvement heatmaps, AHP + TOPSIS, uncertainty criteria, value of time
Policy Comparison	Signals × routing ablation with paired statistics
Simulation Lab	Live re-run of all policies under 11 linked conditions and 3 add-ons; replay; stress test; decision-network pick
Research Extensions	The five extension studies with charts

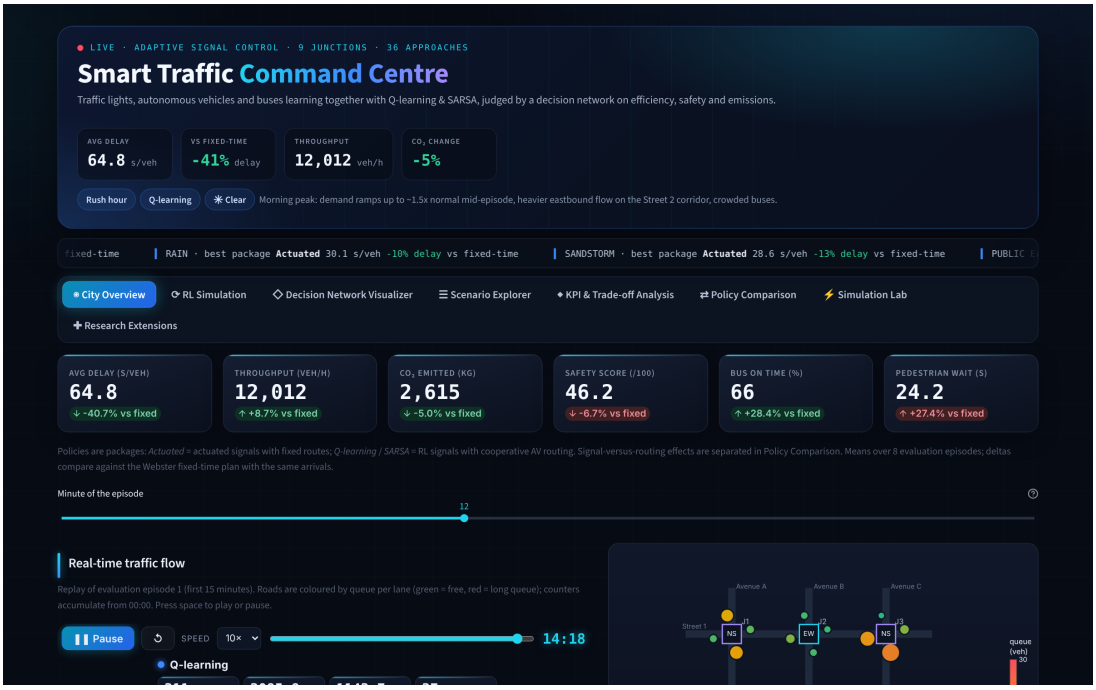


Figure 1. City Overview of the deployed dashboard.

12. Evaluation results and acceptance

Criterion	Result
Delay vs fixed-time (packages)	Q-learning package –41 % in rush hour (64.8 vs 109.3 s) and –49 % in the event surge (46.6 vs 90.6 s); all adaptive controllers within 1-2 s in calm conditions
Signal vs routing effect	Cooperative routing alone –24 % under actuated signals (87.5 → 66.9 s); Q-learning vs actuated signals with the same routing –3 % in rush hour (p = 0.08, not significant)
Fair baseline (corrected)	Against the actuated rule as first written, Q-learning wins the event surge (46.6 vs 62.9 s). With the actuated controller tuned per scenario on validation runs and the same routing, it is a tie (46.9 vs 46.6 s) and tuned actuated is better in rush hour (57.7 vs 64.8 s). RL learns tuned-expert performance without manual tuning but does not beat a well-tuned plan; cooperative routing is the robust gain
Training robustness	Q-learning re-trained three times: 72.5 ± 2.1 s in rush hour

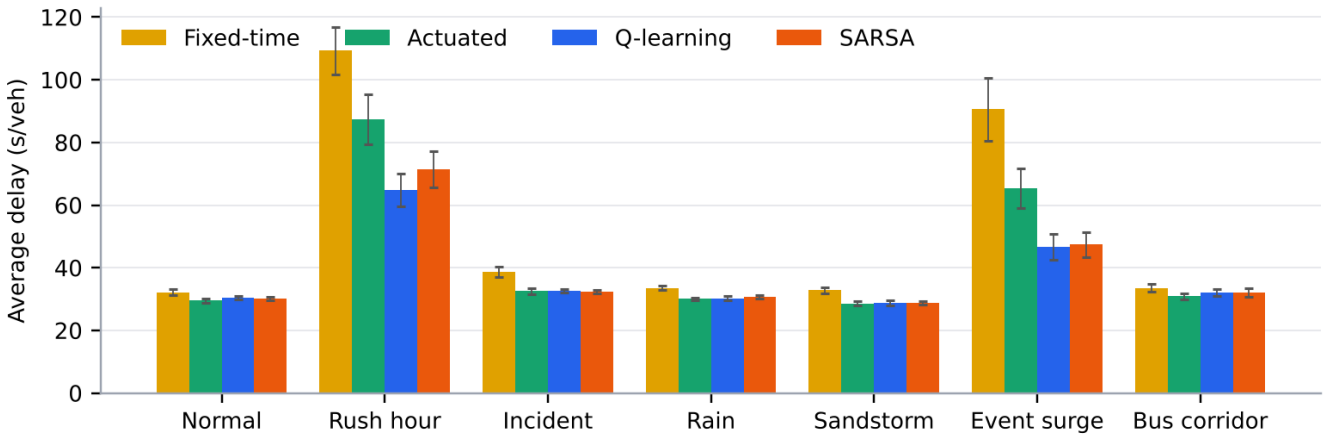


Figure 2. Average delay by scenario and policy package (mean ± s.d., 8 held-out test episodes).

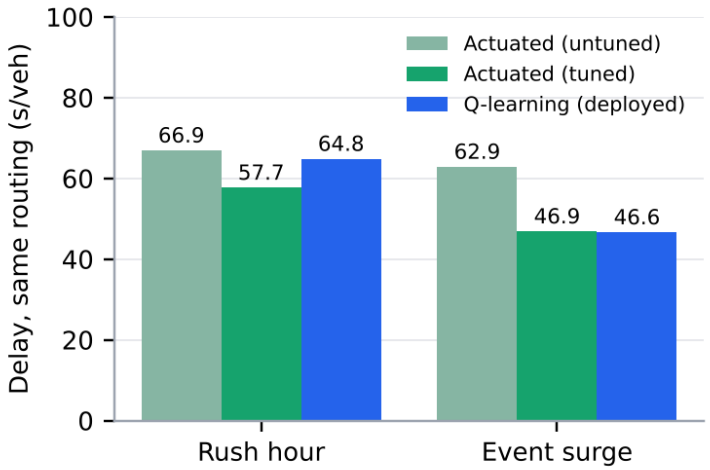


Figure 3. With the same cooperative routing: actuated as first written, actuated tuned on validation runs, and Q-learning.

Research extensions (same deployed agents, same test episodes):

Extension	Finding
AV speed control and lane selection	Lane selection: AV delay –2.2 %, network –0.7 %. Speed advisory: fewer stops, conflicts –1.1 %, CO_2 –0.2 %, for +6.8 % AV delay
Bus load balancing	Rush-hour buses run full under every holding strategy (capacity is binding); a bus every 3 min cuts left-behind passengers from about 35 to 1; RL buses keep the best on-time record (66 % vs 57 %)
Safety-model sensitivity	Halving or doubling each conflict mechanism leaves the safest policy unchanged in 90 % of 42 cases
High-demand safety	Pedestrian max-wait guard (30 s) at 1.6 × demand: conflicts 18.6 → 10.7 per 1,000 veh (actuated 10.6) with delay 179 vs 205 s; switched by demand level

13. Assumptions, constraints and limitations

Synthetic 3 × 3 grid not calibrated to observed Dubai counts; 25 % AV share; expected conflicts are a crash surrogate (rankings robust, absolute values not crash rates); value of time uses a U.S. figure (AED 51.8 per hour, adjustable) and NOx factors are order-of-magnitude; tabular RL limits state detail. Future work: calibrate with RTA detector data, learn the speed advisory, test a larger network.

14. Deliverables

Deliverable	Location
Requirement specification (this document)	Requirement_Specification.pdf
Data CSV	data/smart_traffic_simulation_log.csv (+ supporting CSVs)
Code	Smart_Traffic_MARL_DecisionNetwork.ipynb (Colab) and the Python modules in the repository
Dashboard	https://smarttrafficu-mmzjikh5bg8frby3adzdc2.streamlit.app/
Presentation and report	presentation/Smart_Traffic_Group2.pptx · report/Smart_Traffic_Project_Report.pdf
GenAI prompts	GenAI_disclosure_and_prompts.md